

# APTITUDE QUESTION BANK — VOLUME 2

Deep-Dive, Question-Oriented Master Guide

New Topics + 100+ Fresh Practice Questions with Full Solutions + Full-Length Mock Test

Companion to the Master Aptitude Prep Guide (Volume 1) — this volume adds topics and questions NOT covered there:

Races & Games, Stocks & Shares, Trigonometry, Advanced Mensuration, Installments, Cryptarithmic, Machine Input-Output, Critical Reasoning, Coded Inequalities, Error Spotting, Sentence Improvement, Para Completion, full solved DI sets, and a complete 40-question mock test with answer key.

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# PART 1 — New Quant Topics + Questions

## 1.1 Races & Games (not covered in Vol. 1)

If A gives B a "start" of  $x$  metres in a race of  $d$  metres: A covers  $d$  while B covers  $(d-x)$ . "A beats B by  $t$  seconds" means B needs  $t$  more seconds to finish after A finishes.

**MEDIUM** Q1. In a 100m race, A beats B by 10m and B beats C by 10m. By how many metres does A beat C?

**Solution:** When A runs 100m, B runs 90m. When B runs 100m, C runs 90m  $\rightarrow$  when B runs 90m, C runs  $90 \times 0.9 = 81$ m. So when A runs 100m, C runs 81m. A beats C by  $100 - 81 = 19$  metres.

**HARD** Q2. A can run a km race in 4 min, B in 4 min 40 sec. By what distance does A win the race, assuming A completes the full km?

**Solution:** A takes 240s for 1000m  $\rightarrow$  speed =  $1000/240$  m/s. In the 40s extra B has to run, B covers distance = B's speed  $\times 40$ . B's speed =  $1000/280$  m/s. Distance B covers in 240s =  $(1000/280) \times 240 = 857.14$ m. A wins by  $1000 - 857.14 \approx 142.86$  metres.

## 1.2 Stocks & Shares

Face Value (FV) is fixed; Market Value (MV) fluctuates. Income = (No. of shares)  $\times$  Rate of dividend%  $\times$  FV. Investment = No. of shares  $\times$  MV. Brokerage is added to buy price, subtracted from sell price.

**MEDIUM** Q3. A man buys Rs.100 shares paying 12% dividend at a market value making his yield 15%. Find the market value of each share.

**Solution:** Dividend per share = 12% of 100 = Rs.12. Yield% = (Dividend/MV)  $\times 100 \rightarrow 15 = (12/MV) \times 100 \rightarrow MV = 1200/15 = \text{Rs. } 80$ .

## 1.3 Trigonometry — Heights & Distances

$\tan \theta = \text{Opposite/Adjacent} = \text{Height/Distance}$ . Common values:  $\tan 30^\circ = 1/\sqrt{3}$ ,  $\tan 45^\circ = 1$ ,  $\tan 60^\circ = \sqrt{3}$ . Angle of Elevation = angle above horizontal; Angle of Depression = angle below horizontal (equal to elevation angle from the other point, by alternate angles).

**EASY** Q4. The angle of elevation of the top of a tower from a point 30m away from its base is  $30^\circ$ . Find the height of the tower.

**Solution:**  $\tan 30^\circ = h/30 \rightarrow h = 30 \times (1/\sqrt{3}) = 30/\sqrt{3} = 10\sqrt{3} \approx 17.32$  m.

**HARD** Q5. From the top of a 60m tower, the angle of depression of a car is  $30^\circ$ . How far is the car from the base of the tower?

**Solution:** Angle of depression = angle of elevation from car =  $30^\circ$ .  $\tan 30^\circ = 60/d \rightarrow d = 60/\tan 30^\circ = 60 \times \sqrt{3} = 60 \times 1.732 \approx 103.9$  m.

## 1.4 Advanced Mensuration — Frustum & Combined Solids

Frustum of a cone (radii  $R, r$ ; height  $h$ ; slant  $l = \sqrt{h^2 + (R-r)^2}$ ): Volume =  $(1/3)\pi h(R^2 + r^2 + Rr)$ . CSA =  $\pi l(R+r)$ . TSA =  $\pi l(R+r) + \pi R^2 + \pi r^2$

**HARD** Q6. A bucket in the shape of a frustum has radii 15cm and 5cm and height 24cm. Find its volume. ( $\pi = 3.14$ )

**Solution:** Vol =  $(1/3) \times 3.14 \times 24 \times (225 + 25 + 75) = (1/3) \times 3.14 \times 24 \times 325 = (1/3) \times 24492 = 8164 \text{ cm}^3$  (approx).

## 1.5 Set Theory — Three-Set Problems (deeper)

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$$

**HARD** Q7. In a survey of 100 students: 45 like Tea, 35 like Coffee, 40 like Juice, 10 like Tea & Coffee, 15 like Coffee & Juice, 12 like Tea & Juice, 5 like all three. How many like none?

**Solution:** At least one =  $45 + 35 + 40 - 10 - 15 - 12 + 5 = 88$ . None =  $100 - 88 = 12$  students.

## 1.6 Compound Interest with Installments

**Key Point:** For a debt repaid in  $n$  equal annual installments (amount  $X$ , at  $r\%$  CI), each installment must account for accrued interest on the remaining principal. Standard equation:  $\text{Debt} = X/(1+r/100) + X/(1+r/100)^2 + \dots + X/(1+r/100)^n$ .

**HARD** Q8. A sum of Rs.2100 is to be repaid in 2 equal annual installments at 10% CI. Find the installment amount.

**Solution:** Let installment =  $X$ .  $2100 = X/1.1 + X/1.21$ . Multiply through by 1.21:  $2100 \times 1.21 = 1.1X + X \rightarrow 2541 = 2.1X \rightarrow X = \text{Rs. } 1210$ .

## 1.7 Surds & Indices — Extra Practice

**MEDIUM** Q9. Simplify:  $\sqrt{12 + \sqrt{12 + \sqrt{12 + \dots \infty}}}$

**Solution:** Let  $x = \sqrt{12 + x} \rightarrow x^2 = 12 + x \rightarrow x^2 - x - 12 = 0 \rightarrow (x - 4)(x + 3) = 0 \rightarrow x = 4$  (rejecting negative). **Answer: 4**

**MEDIUM** Q10. If  $2^x = 3^y = 6^{-z}$ , show the relation between  $x, y, z$  (find value of  $1/x + 1/y + 1/z$ ).

**Solution:** Let  $2^x = 3^y = 6^{-z} = k$ . Then  $2 = k^{1/x}$ ,  $3 = k^{1/y}$ ,  $6 = k^{-1/z}$ . Since  $2 \times 3 = 6$ :  $k^{1/x} \times k^{1/y} = k^{-1/z} \rightarrow 1/x + 1/y = -1/z \rightarrow 1/x + 1/y + 1/z = 0$ .

## 1.8 Approximation & Fast Estimation

**Key Point:** For "approximate value" questions, round each number to 1-2 significant figures before computing, then adjust. E.g., 48.98% of 649.97 + 15.02% of 320.05  $\approx$  49% of 650 + 15% of 320 = 318.5 + 48 = 366.5.

**EASY** Q11. Find the approximate value of:  $6799 \div 199.8 \times 4.02$

**Solution:**  $\approx 6800/200 \times 4 = 34 \times 4 = 136$ .

## PART 2 — New Reasoning Topics + Questions

### 2.1 Machine Input-Output (very common in TCS/IBPS, not covered in Vol.1)

**Key Point:** A word/number arrangement is processed step-by-step by a "machine" following a hidden rule (e.g., shifting largest number to front, arranging words alphabetically, etc.). Study each step's change from the previous step to deduce the rule, then apply it to find an intermediate/final step for a new input.

**HARD** Q12. Input: 64 25 9 81 36 49. Step I: 9 64 25 81 36 49. Step II: 9 25 64 81 36 49. Step III: 9 25 36 64 81 49. Step IV: 9 25 36 49 64 81. What is the rule, and what is Step V (if any)?

**Solution:** Each step, the next-smallest remaining number is moved to its correct sorted position from the left (essentially a selection-sort, one pass per step). By Step IV the list is already fully sorted ascending (9,25,36,49,64,81), so **no Step V is needed — sorting is already complete.**

### 2.2 Cryptarithmic

**Key Point:** Each letter represents a unique digit (0-9). Start from the column with the fewest unknowns (usually units or the leftmost carry column). Use carry-over logic systematically.

**HARD** Q13. Solve: SEND + MORE = MONEY (classic cryptarithmic). What is the value of M?

**Solution:** This is the famous unique solution: S=9, E=5, N=6, D=7, M=1, O=0, R=8, Y=2 → 9567+1085=10652. **M = 1** (M must be 1 since the 5-digit sum MONEY starting with M means M is the carry-over from a 4-digit + 4-digit addition, which can only be 1).

### 2.3 Critical Reasoning — Statement & Argument / Strong-Weak Arguments

**Key Point:** A "strong" argument must be both important/relevant to the issue AND factually/logically valid. A "weak" argument may be true but irrelevant, too general, or based on assumption rather than fact.

**MEDIUM** Q14. Statement: Should all public transport in cities be made free? Argument I: Yes, it will reduce traffic congestion and pollution significantly. Argument II: No, because buses are usually painted blue.

**Solution:** Argument I is **strong** — directly relevant with a plausible causal link. Argument II is **weak** — irrelevant to the cost/free debate. **Answer: Only Argument I is strong.**

### 2.4 Coded Inequalities / Symbols & Notations

**Key Point:** Symbols replace mathematical signs, e.g., 'A@B' might mean 'A>B', 'A#B' means 'A≥B', etc. Convert the entire coded statement to normal inequality form first, then combine using transitive chains ( $A>B>C \Rightarrow A>C$ ) before checking the conclusions.

**MEDIUM** Q15. If 'P\$Q' means  $P>Q$ , 'P&Q' means  $P<Q$ , 'P#Q' means  $P=Q$ . Statements: A\$B, B&C, C#D. Conclusion: A\$D?

**Solution:**  $A>B$ ,  $B<C$ ,  $C=D$ . From  $A>B$  we cannot directly relate A and C ( $B<C$  doesn't tell us if A is bigger or smaller than C). **Conclusion A\$D ( $A>D$ ) does not necessarily follow — relationship is indeterminate.**

### 2.5 Ranking & Order (Linear Ranking, not covered in Vol.1)

**Key Point:** If a person's rank from the top is x and total people is n, rank from bottom =  $n-x+1$ . Careful with "just above/below" (adjacent) vs "before/after" (relative order, not necessarily adjacent).

**EASY** Q16. In a class of 40 students, Rahul ranks 17th from the top. What is his rank from the bottom?

**Solution:** Rank from bottom =  $40 - 17 + 1 = 24\text{th}$ .

**MEDIUM** Q17. In a row of children facing North, Meena is 7th from the left and 12th from the right. How many children are in the row?

**Solution:** Total = (Position from left) + (Position from right) - 1 =  $7 + 12 - 1 = 18$  children.

## 2.6 Mixed & Alpha-Numeric Series (extra variety)

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**MEDIUM** Q18. Find the odd one: 8, 27, 64, 125, 196, 216

**Solution:** These are cubes:  $2^3=8$ ,  $3^3=27$ ,  $4^3=64$ ,  $5^3=125$ ,  $6^3=216$  ✓ but  $196=14^2$  (a square, not a cube). **Odd one: 196.**

**HARD** Q19. Find the next term: Z, X, U, Q, ?

**Solution:** Position from Z(26): Z=26, X=24(-2), U=21(-3), Q=17(-4), next subtract 5 →  $17 - 5 = 12 = \text{L}$ . **Answer: L.**

## 2.7 Word Formation & Dictionary Order (new sub-topic)

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**EASY** Q20. Arrange the words as per dictionary order and find which comes 3rd: (1) Mountain (2) Moth (3) Mother (4) Motion (5) Mound

**Solution:** Dictionary order compares letter by letter: Motion(4), Moth(2), Mother(3), Mound(5), Mountain(1). Sorted: Moth, Mother, Motion, Mound, Mountain → **3rd = Motion.**

## PART 3 — New Verbal Topics + Questions

### 3.1 Spotting Errors (Error Detection — not covered in Vol.1)

**Key Point:** Sentence is split into parts (A)/(B)/(C)/(D); find the part with a grammatical error, or mark "No error". Focus on: subject-verb agreement, tense, preposition, article, and countable/uncountable noun usage.

**MEDIUM** Q21. (A) Neither of the two boys / (B) have done / (C) their homework / (D) on time. / No error

**Solution:** "Neither" is singular, so verb must be "has", not "have". **Error in part (B)** — should be "has done".

**MEDIUM** Q22. (A) The number of students / (B) in the class are / (C) increasing / (D) every year. / No error

**Solution:** "The number of" takes a singular verb ("is"), not "are". **Error in part (B)** — should be "is increasing".

### 3.2 Sentence Improvement

**Key Point:** Identify the underlined/error segment, then pick the option that fixes the grammar/idiom WITHOUT changing the intended meaning. Avoid options that are grammatically correct but change the meaning.

**MEDIUM** Q23. Improve: "He is one of the best player in the team."

**Solution:** "One of the best" must be followed by a plural noun. **Correction:** "He is one of the best players in the team."

### 3.3 Para Completion (choosing the concluding sentence)

**Key Point:** The correct concluding sentence should logically continue the tone, tense, and topic of the paragraph — not introduce a new unrelated idea. Check for continuity of pronouns/subjects from the last given sentence.

**HARD** Q24. Paragraph: "Renewable energy adoption has accelerated globally over the past decade. Solar and wind costs have fallen sharply, making them competitive with fossil fuels. \_\_\_" Which sentence best completes the paragraph? (a) Coal mining remains dangerous work. (b) As a result, many countries are now revising their energy policies to prioritize clean power. (c) The moon orbits the Earth every 27 days. (d) Renewable jobs require no training.

**Solution:** Must logically follow from "costs falling → competitive" as a consequence. **Answer: (b)** — direct logical continuation ("As a result...").

### 3.4 Double Fill-in-the-Blanks (Sentence Completion)

**MEDIUM** Q25. "Despite the team's \_\_\_ efforts, the project was ultimately \_\_\_ due to lack of funding." (a) diligent, abandoned (b) careless, completed (c) diligent, celebrated (d) poor, funded

**Solution:** "Despite" signals contrast — strong effort but a negative outcome. **Answer: (a) diligent, abandoned.**

### 3.5 Extended Vocabulary — Root Words (new list)

| Root   | Meaning | Example                 |
|--------|---------|-------------------------|
| chron- | time    | chronology, synchronize |
| bio-   | life    | biology, biography      |
| phon-  | sound   | phonetics, telephone    |
|        |         |                         |

|               |            |                         |
|---------------|------------|-------------------------|
| graph-        | write      | autograph, biography    |
| vid/vis-      | see        | visible, television     |
| ject-         | throw      | reject, project         |
| duc/duct-     | lead       | conduct, induce         |
| scrib/script- | write      | describe, manuscript    |
| omni-         | all        | omnipresent, omniscient |
| mis-/miso-    | hate/wrong | misanthrope, misconduct |

### 3.6 Formal Email / Business Communication Basics (Accenture/Capgemini communication rounds)

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**Key Point:** Structure: Greeting → Purpose (1st line) → Body (2-3 short paras) → Action requested → Polite closing. Keep tone formal, avoid contractions ("don't" → "do not"), avoid overly casual words, proofread for subject-verb agreement and tense consistency before submitting.



## PART 4 — Fully Solved Data Interpretation Sets

### 4.1 Table-Based DI Set

The table shows the number of employees (in hundreds) in five departments of a company over two years.

| Department | 2023 | 2024 |
|------------|------|------|
| HR         | 12   | 15   |
| Sales      | 30   | 36   |
| IT         | 45   | 54   |
| Finance    | 20   | 22   |
| Ops        | 25   | 28   |

**EASY** Q26. What is the % increase in IT department employees from 2023 to 2024?

**Solution:** Increase =  $54 - 45 = 9$ . % =  $(9/45) \times 100 = 20\%$ .

**MEDIUM** Q27. What is the ratio of total employees in 2023 to total employees in 2024?

**Solution:** 2023 total =  $12 + 30 + 45 + 20 + 25 = 132$ . 2024 total =  $15 + 36 + 54 + 22 + 28 = 155$ . Ratio = 132:155 (already in lowest terms, GCD=1... check:  $\text{GCD}(132, 155) = 1$ ) → **132:155**.

**HARD** Q28. Which department had the highest percentage growth?

**Solution:** HR:  $3/12 = 25\%$ . Sales:  $6/30 = 20\%$ . IT:  $9/45 = 20\%$ . Finance:  $2/20 = 10\%$ . Ops:  $3/25 = 12\%$ . Highest = **HR at 25%**.

### 4.2 Pie Chart DI Set

A pie chart shows the distribution of a family's monthly budget of Rs. 60,000: Rent 30%, Food 25%, Savings 20%, Education 15%, Others 10%.

**EASY** Q29. How much is spent on Food?

**Solution:** 25% of 60000 = **Rs. 15,000**.

**MEDIUM** Q30. What is the central angle for the "Savings" sector?

**Solution:**  $20\% \times 360^\circ = 72^\circ$ .

**MEDIUM** Q31. How much more is spent on Rent than on Education?

**Solution:** Rent =  $30\% \times 60000 = 18000$ . Education =  $15\% \times 60000 = 9000$ . Difference = **Rs. 9,000**.

### 4.3 Line Graph DI Set

A line graph shows a company's monthly revenue (in Rs. lakh): Jan 40, Feb 55, Mar 50, Apr 65, May 70, Jun 60.

**EASY** Q32. Between which two consecutive months was the increase in revenue the highest?

**Solution:** Changes: Jan-Feb +15, Feb-Mar -5, Mar-Apr +15, Apr-May +5, May-Jun -10. Highest increase tie between Jan-Feb and Mar-Apr, both +15. **Answer: Jan-Feb and Mar-Apr (tie, +15 lakh each)**.

**MEDIUM** Q33. Find the average monthly revenue over the 6 months.

**Solution:** Sum =  $40 + 55 + 50 + 65 + 70 + 60 = 340$ . Average =  $340/6 = 56.67$  lakh.

### 4.4 Caselet DI Set

Caselet: "In a survey of 800 people about their preferred mode of commute, 40% preferred Car, 25% preferred Public Transport, 20% preferred Bike, and the rest preferred Cycling. Among Car users, 60% were men."

**MEDIUM** Q34. How many people preferred Cycling?

**Solution:** Remaining % =  $100 - 40 - 25 - 20 = 15\%$ . 15% of 800 = **120 people**.

**HARD** Q35. How many women preferred Car?

**Solution:** Car users =  $40\% \times 800 = 320$ . Men =  $60\% \times 320 = 192$ . Women =  $320 - 192 = \mathbf{128}$ .

## PART 5 — Extra Practice Bank: Quantitative Aptitude (25 New Questions)

**EASY** Q36. Find the unit digit of  $3^{57}$ .

**Sol:** Cycle(3)=3,9,7,1.  $57 \bmod 4 = 1 \rightarrow$  1st term=3. **Answer: 3**

**EASY** Q37. What is the LCM of 18, 24, and 36?

**Sol:**  $18=2 \times 3^2$ ,  $24=2^3 \times 3$ ,  $36=2^2 \times 3^2$ .  $\text{LCM}=2^3 \times 3^2=8 \times 9=72$

**MEDIUM** Q38. A number is increased by 30% and then decreased by 30%. What is the net % change?

**Sol:**  $\text{Net}=30-30+(30 \times -30/100)=-9\%$ . **Answer: 9% decrease**

**MEDIUM** Q39. A shopkeeper sells an article at a loss of 15%. Had he sold it for Rs.51 more, he would have gained 5%. Find the cost price.

**Sol:**  $20\% \text{ of CP} = 51 \rightarrow \text{CP}=51/0.20=\text{Rs. } 255$

**HARD** Q40. At what rate percent per annum will Rs.3000 amount to Rs.3993 in 3 years at compound interest?

**Sol:**  $(1+r/100)^3 = 3993/3000 = 1.331 = 1.1^3 \rightarrow 1+r/100=1.1 \rightarrow r=10\%$

**EASY** Q41. Divide Rs.4590 among A, B, C so that A gets  $\frac{2}{3}$  of what B gets and B gets  $\frac{1}{4}$  of what C gets. Find C's share.

**Sol:** Let  $C=12k$ ,  $B=(\frac{1}{4}) \times 12k=3k$ ,  $A=(\frac{2}{3}) \times 3k=2k$ . Ratio A:B:C=2:3:12, total parts=17.  $17k=4590 \rightarrow k=270$ . C's share= $12k=\text{Rs. } 3240$

**MEDIUM** Q42. The average weight of 8 persons increases by 2.5kg when a new person replaces one weighing 65kg. Find the weight of the new person.

**Sol:** Increase in total =  $8 \times 2.5=20$ . New weight= $65+20=85 \text{ kg}$

**EASY** Q43. The sum of the ages of a father and son is 60 years. Six years ago father's age was 5 times son's age. Find son's present age.

**Sol:**  $F+S=60$ .  $(F-6)=5(S-6) \rightarrow F=5S-24$ . Sub:  $5S-24+S=60 \rightarrow 6S=84 \rightarrow S=14 \text{ years}$

**HARD** Q44. A, B and C can do a piece of work in 10, 15 and 30 days respectively. If A is assisted by B and C on alternate days, how long will A take to finish the work?

**Sol:** Day1(A+B)= $\frac{1}{10}+\frac{1}{15}=\frac{1}{6}$ . Day2(A+C)= $\frac{1}{10}+\frac{1}{30}=\frac{2}{15}$ . 2-day cycle= $\frac{1}{6}+\frac{2}{15}=\frac{5}{30}+\frac{4}{30}=\frac{9}{30}=\frac{3}{10}$ . In 6 days(3 cycles)= $\frac{9}{10}$  done. Remaining= $\frac{1}{10}$  on Day7 (A+B day)= $\frac{1}{6}$  per day > needed  $\frac{1}{10}$ , so finishes within day 7: time= $(\frac{1}{10})/(\frac{1}{6})=0.6 \text{ day}$ . Total= $6+0.6=6.6 \text{ days}$

**MEDIUM** Q45. A car covers a distance of 540 km in 9 hours. What is the speed in m/s?

**Sol:** Speed= $540/9=60 \text{ km/hr}=60 \times \frac{5}{18}=16.67 \text{ m/s}$

**HARD** Q46. Two trains start at the same time from stations A and B towards each other and after crossing they take 4 and 9 hours to reach B and A respectively. If the speed of the first train is 60 km/hr, find the speed of the second train.

**Sol:** Formula:  $(\text{Speed1}/\text{Speed2})=\sqrt{(t_2/t_1)}=\sqrt{(9/4)}=3/2$ . Speed2= $60 \times (2/3)=40 \text{ km/hr}$

**MEDIUM** Q47. In what ratio should tea worth Rs.60/kg be mixed with tea worth Rs.90/kg so that the mixture is worth Rs.80/kg?

**Sol:** Ratio= $(90-80):(80-60)=10:20=1:2$

**EASY** Q48. How many ways can a committee of 3 be chosen from 7 people?

**Sol:**  ${}^7C_3 = 7!/(3!4!) = 35$  ways

**HARD** Q49. A bag has 4 red and 6 blue balls. Two balls are drawn at random. Find the probability both are blue.

**Sol:** Total ways =  ${}^{10}C_2 = 45$ . Favourable =  ${}^6C_2 = 15$ .  $P = 15/45 = 1/3$

**MEDIUM** Q50. Find the sum of the first 15 terms of the GP: 3, 6, 12, 24...

**Sol:**  $a=3, r=2, n=15$ . Sum =  $a(r^n - 1)/(r - 1) = 3(2^{15} - 1)/1 = 3 \times 32767 = 98301$

**EASY** Q51. Solve for x:  $3x - 7 = 2x + 5$

**Sol:**  $3x - 2x = 5 + 7 \rightarrow x = 12$

**HARD** Q52. The perimeter of a rectangle is 100m and its length is 10m more than its breadth. Find its area.

**Sol:**  $2(l+b) = 100 \rightarrow l+b = 50$ .  $l = b + 10 \rightarrow 2b + 10 = 50 \rightarrow b = 20, l = 30$ . Area = **600 m<sup>2</sup>**

**MEDIUM** Q53. Find the total surface area of a hemisphere with radius 7cm. ( $\pi = 22/7$ )

**Sol:** TSA =  $3\pi r^2 = 3 \times (22/7) \times 49 = 462$  cm<sup>2</sup>

**EASY** Q54. What is 15% of 15% of 4000?

**Sol:** 15% of 4000 = 600. 15% of 600 = **90**

**HARD** Q55. A sum was put at SI at a certain rate for 3 years. Had it been put at a rate 2% higher, it would have fetched Rs.360 more. Find the sum.

**Sol:** Extra SI =  $P \times 2 \times 3/100 = 360 \rightarrow P = 360 \times 100/6 = \text{Rs. } 6000$

**MEDIUM** Q56. Find the value of x if  $2^{(x+1)} = 32$ .

**Sol:**  $32 = 2^5$ , so  $x+1 = 5 \rightarrow x = 4$

**MEDIUM** Q57. A man spends 75% of his income. His income increases by 20% and his expenditure also increases by 10%. Find the % increase in his savings.

**Sol:** Let income = 100. Old exp = 75, old savings = 25. New income = 120, new exp = 82.5, new savings = 37.5. Increase =  $(37.5 - 25)/25 \times 100 = 50\%$

**HARD** Q58. A container has milk and water in ratio 5:1. If 12 litres of the mixture is replaced with water, the ratio becomes 5:3. Find the original quantity of mixture.

**Sol:** Milk fraction removed unchanged formula: let total = V. Milk left after removing 12L mixture and adding 12L water =  $(5/6)V \times (1 - 12/V) \dots$  Using shortcut: initial milk fraction 5/6, final milk fraction 5/8.  $(5/6)(1 - 12/V) = 5/8 \rightarrow 1 - 12/V = (5/8)/(5/6) = 6/8 = 3/4 \rightarrow 12/V = 1/4 \rightarrow V = 48$  litres

**EASY** Q59. Find the median of: 12, 18, 9, 21, 15

**Sol:** Sorted: 9, 12, 15, 18, 21. Median (middle value) = **15**

**MEDIUM** Q60. A and B together can complete a task in 6 days. A alone can do it in 10 days. In how many days can B alone complete it?

**Sol:** B's rate =  $1/6 - 1/10 = 5/30 - 3/30 = 2/30 = 1/15$ . B alone = **15 days**

## PART 6 — Extra Practice Bank: Logical Reasoning (20 New Questions)

**EASY** Q61. Complete the series: A, C, F, J, O, ?

**Sol:** Gaps increase by 1: +2,+3,+4,+5,+6.  $O(15)+6=21$ st letter=**U**

**MEDIUM** Q62. If in a code language each letter of a word is replaced by the next letter of the alphabet and the resulting word is then reversed, how would "FLOWER" be coded?

**Sol:** Step 1 — shift each letter +1: F→G, L→M, O→P, W→X, E→F, R→S, giving GMPXFS. Step 2 — reverse the whole string: **SFXPMG**. **Key takeaway:** always apply the transformation rule exactly as defined in the question (here: shift, then reverse) rather than assuming a different fixed pattern.

**EASY** Q63. Pointing to a man, a woman says, "He is the son of my husband's mother." Who is the man to the woman?

**Sol:** Son of husband's mother = husband's brother OR husband himself. Standard answer (excluding self): **Brother-in-law (husband's brother)**

**MEDIUM** Q64. A man walks 4km South, turns right and walks 3km, then turns right again and walks 4km. How far is he from the starting point?

**Sol:** South 4, then West→ turning right while facing South = West, 3km. Then right again (facing North) 4km — net: South-North cancels ( $4-4=0$ ), only West 3km remains. Distance=**3 km**

**HARD** Q65. Statements: Some cats are dogs. All dogs are rabbits. Conclusions: I. Some cats are rabbits. II. Some rabbits are dogs.

**Sol:** "Some cats are dogs" + "All dogs are rabbits" → Some cats are rabbits (valid syllogism chain) — I follows. "All dogs are rabbits" converts to "Some rabbits are dogs" — II follows. **Answer: Both I and II follow**

**MEDIUM** Q66. Six friends P,Q,R,S,T,U sit around a circular table facing the centre. P sits between Q and R. S sits opposite to P. T is to the immediate left of Q. Where does U sit?

**Sol:** Arranging: Q-P-R around one side, S opposite P (across the table), T immediate left of Q. By elimination, the only remaining seat is between S and R (or S and Q depending on exact layout) — **U occupies the one remaining seat, adjacent to S** (solve by drawing the 6-seat circle and placing each clue in order).

**EASY** Q67. If SUN is coded as 45 and MOON is coded as 60 (sum of letter positions), what is the code for STAR?

**Sol:** S=19,T=20,A=1,R=18. Sum= $19+20+1+18=58$

**MEDIUM** Q68. Find the missing number in the matrix: Row: 4,9,25 → 2,3,5 (square root pattern); Row: 16,49,? → 4,7,11

**Sol:** Pattern is square of the number:  $4^2=16$ ,  $7^2=49$ ,  $11^2=121$

**HARD** Q69. In a certain code, "246" means "roses are red", "637" means "red and blue", "829" means "blue is nice". What digit means "red"?

**Sol:** "246"=roses are red, "637"=red and blue. Common word "red" ↔ common digit between 246 and 637 = **6**

**MEDIUM** Q70. Assertion (A): All prime numbers greater than 2 are odd. Reason (R): 2 is the only even prime number. Choose: both true and R explains A / both true but R doesn't explain A / A true R false / A false R true.

**Sol:** Both statements are factually true, and R (2 being the only even prime) directly explains why all other primes must be odd. **Answer: Both A and R are true, and R is the correct explanation of A**

**EASY** Q71. Which figure completes the analogy: Book:Library :: Fish:? (word analogy, non-visual)

**Sol:** Book is kept/found in a Library. Fish is found in an **Aquarium/Ocean** (context-dependent; typical MCQ answer: Aquarium)

**MEDIUM** Q72. Today is Wednesday. What day will it be 100 days from now?

**Sol:**  $100 \bmod 7 = 2$ . Wednesday+2 days = **Friday**

**HARD** Q73. A dice has numbers 1-6. In three positions: (1,2,3 visible), (2,4,5 visible), (1,5,6 visible). Which number is opposite to 2?

**Sol:** In position1, 2 appears with 1,3 (not opposite). In position2, 2 appears with 4,5. So 2 is never seen with 6 across all throws → 2 is opposite **6**

**MEDIUM** Q74. Statement: "All employees must submit their reports by Friday." Assumption: Employees are aware of what a 'report' means in this context.

**Sol:** This is a valid implicit assumption — the instruction only works if employees understand what "report" refers to. **Assumption is implicit and valid.**

**EASY** Q75. Odd one out: Triangle, Square, Circle, Perimeter

**Sol:** First three are shapes; "Perimeter" is a measurement, not a shape. **Odd one: Perimeter**

**MEDIUM** Q76. P is the brother of Q. R is the mother of P. S is the father of Q. If R and S are married, how is S related to P?

**Sol:** S is father of Q, and R (P's mother) is married to S → S is **Father of P**

**HARD** Q77. Five books P,Q,R,S,T are placed on a shelf. P is to the left of Q but right of R. S is to the right of T but left of R. Arrange them left to right.

**Sol:** From clues: T-S-R-P-Q (T leftmost since S is right of T, S left of R, R left of P, P left of Q). **Order: T, S, R, P, Q**

**MEDIUM** Q78. If '+' means '÷', '-' means '×', '×' means '+', '÷' means '-', evaluate:  $8 \times 4 \div 2 + 4 - 3$

**Sol:** Replace symbols:  $8+4-2\div 4\times 3 = 8+4-0.5\times 3=8+4-1.5=10.5$  (apply BODMAS after substitution)

**EASY** Q79. Find the number of triangles in a standard 5-pointed star (Pentagram) figure — a commonly asked non-verbal fact.

**Sol:** A standard pentagram figure contains **10 triangles** (5 small point-triangles + 5 larger ones formed by combining adjacent points) — memorize this as a standard figure-counting fact.

**HARD** Q80. In a queue, Anil is 10th from the front and Bala is 15th from the end. If there are 5 people between Anil and Bala, find the total number of people in the queue (two possible arrangements exist — give both).

**Sol:** Case 1 (Bala behind Anil): Total = Anil's position + 5 (gap) + Bala's position from end =  $10+5+15=30$ . Case 2 (Anil behind Bala) generally isn't valid here since Anil is defined from front and Bala from end consistently — standard answer: **30 people**.

## PART 7 — Extra Practice Bank: Verbal Ability (15 New Questions)

**EASY** Q81. Choose the synonym of "Meticulous": (a) Careless (b) Precise (c) Slow (d) Loud

**Sol:** Meticulous = showing great attention to detail. **Answer: (b) Precise**

**EASY** Q82. Choose the antonym of "Benevolent": (a) Kind (b) Generous (c) Malevolent (d) Happy

**Sol:** Benevolent=kind-hearted; opposite=(c) Malevolent

**MEDIUM** Q83. Fill in the blank: "She is not only intelligent \_\_\_ also hardworking."

**Sol:** Correlative pair "not only...but also". **Answer: but**

**MEDIUM** Q84. Choose the correctly spelt word: (a) Accomodate (b) Accommodate (c) Accommodate (d) Acomodate

**Sol:** Answer: (c) Accommodate (double 'c', double 'm')

**HARD** Q85. Rearrange to form a meaningful sentence: P) had studied hard Q) but R) he still failed S) for the exam

**Sol:** Logical order: P-S-Q-R = "He had studied hard for the exam but he still failed." **Answer: PSQR**

**EASY** Q86. One word substitution: "A person who is difficult to please" is called: (a) Fastidious (b) Amiable (c) Gracious (d) Docile

**Sol:** Answer: (a) Fastidious

**MEDIUM** Q87. Choose the correct preposition: "He is good \_\_\_ mathematics."

**Sol:** "good at" is the correct idiomatic usage. **Answer: at**

**HARD** Q88. Identify the error: "Each of the students have submitted their assignment."

**Sol:** "Each" is singular → verb should be "has" not "have". **Error: "have" should be "has"**

**MEDIUM** Q89. Choose the idiom meaning "to reveal a secret": (a) Spill the beans (b) Hit the sack (c) Under the weather (d) Kick the bucket

**Sol:** Answer: (a) Spill the beans

**EASY** Q90. Fill in the blank: "The train \_\_\_ the station five minutes ago." (a) leaves (b) has left (c) left (d) is leaving

**Sol:** A specific past time marker ("five minutes ago") requires simple past. **Answer: (c) left**

**MEDIUM** Q91. Choose the word closest in meaning to "Ephemeral": (a) Permanent (b) Fleeting (c) Ancient (d) Solid

**Sol:** Ephemeral=lasting a very short time. **Answer: (b) Fleeting**

**HARD** Q92. Rearrange the parts (para jumble): 1) The internet has transformed 2) it possible to access information instantly 3) how we communicate 4) making

**Sol:** Correct order: 1-3-4-2: "The internet has transformed how we communicate, making it possible to access information instantly." **Answer: 1,3,4,2**

**MEDIUM** Q93. Choose the correct passive voice: "They are building a new bridge."

**Sol:** Answer: "A new bridge is being built (by them)."

**EASY** Q94. Antonym of "Verbose": (a) Wordy (b) Concise (c) Talkative (d) Lengthy

**Sol:** Verbose=using more words than needed. Opposite=(b) Concise

**HARD** Q95. Reading comprehension inference-type: If a passage states "Despite record profits, the company announced layoffs to cut costs," which of these is the best inference? (a) The company is going bankrupt (b) The layoffs indicate cost-cutting priorities independent of current profitability (c) Employees caused the profit loss (d) The company will shut down next year

**Sol:** Only (b) is directly and safely inferable from the statement without over-extrapolating. **Answer: (b)**



## PART 8 — Full-Length Mixed Mock Test (40 Questions)

Simulate real test conditions: 40 minutes, no pausing. Attempt all, then check Part 9 for the answer key and explanations. Sections: Q1-15 Quant, Q16-30 Reasoning, Q31-40 Verbal.

### Section I: Quantitative Aptitude (Q1-Q15)

- Find the unit digit of  $6^{233} \times 7^{92}$ .
- The HCF of two numbers is 12 and their LCM is 336. If one number is 48, find the other.
- A sum becomes Rs.5,700 in 3 years and Rs.6,300 in 5 years at SI. Find the rate of interest.
- A shopkeeper allows a 10% discount and still makes a 20% profit. If the cost price is Rs.500, find the marked price.
- The ratio of ages of A and B is 3:5. After 9 years, the ratio becomes 3:4. Find A's present age.
- 15 workers can complete a job in 24 days. How many workers are needed to complete it in 18 days?
- A boat's speed in still water is 12 km/hr and the stream's speed is 4 km/hr. Find the time to cover 32 km downstream.
- In how many ways can 5 different books be arranged on a shelf?
- A bag contains 5 red and 3 green balls. One ball is drawn at random. Find the probability it is green.
- Find the 10th term of the AP: 2, 5, 8, 11...
- Find the area of a triangle with sides 9, 12, 15.
- If  $x:y = 3:4$ , find  $(2x+3y):(3x-y)$ .
- The average of 7 numbers is 24. If one number is excluded, the average becomes 21. Find the excluded number.
- Simplify:  $(0.75 \times 0.75 \times 0.75 - 0.001)/(0.75 \times 0.75 + 0.75 \times 0.1 + 0.01)$
- A sum of Rs.8000 amounts to Rs.9261 in 3 years at compound interest. Find the rate.

### Section II: Logical Reasoning (Q16-Q30)

- Find the next term: 1, 4, 9, 16, 25, ?
- If in a code, TABLE is written as UBCMF, what is CHAIR written as?
- Pointing to a photograph, Rita said, "He is the son of my father's only daughter." How is the boy related to Rita?
- A man walks 6 km East, then 8 km North. Find his distance from the starting point.
- Statements: All squares are rectangles. All rectangles are quadrilaterals. Conclusion: All squares are quadrilaterals — valid or invalid?
- Six people sit in a row. D is to the right of E. F is to the left of E but right of G. Who is at the leftmost position (given only these 4 people, in relative order)?
- If 'A' is coded as 1, 'B' as 2, ... 'Z' as 26, find the code for the word "CAB".
- How many Sundays are there in a month starting on a Sunday and having 31 days?
- A is the father of B. C is the sister of A. D is the mother of C. How is D related to B?
- Odd one out: 121, 144, 169, 200, 225
- Find the missing number: 7, 14, 28, 56, ?
- In a certain code, PAPER is written as QBQFS. How is PENCIL written in that code?
- If Monday falls on the 1st of a month, what day is the 29th?
- Statement: Should smoking be banned in all public places? Argument: Yes, it protects non-smokers from second-hand smoke health risks. Strong or weak?
- Ravi ranks 9th from the top and 26th from the bottom in a class. How many students are in the class?

### Section III: Verbal Ability (Q31-Q40)

- Synonym of "Candid": (a) Dishonest (b) Frank (c) Secretive (d) Confused
- Antonym of "Diligent": (a) Hardworking (b) Lazy (c) Careful (d) Sincere
- Fill in the blank: "Neither the manager nor the employees \_\_\_ satisfied with the outcome."
- Choose the correctly spelt word: (a) Recieve (b) Receive (c) Recieve (d) Receve
- Identify the error: "One of my friend are getting married next week."
- One word substitution: "A person who can speak two languages" is: (a) Bilingual (b) Multilingual (c) Linguist (d) Translator
- Idiom meaning: "To beat around the bush" means: (a) Speak directly (b) Avoid the main topic (c) Get angry (d) Work hard
- Choose the correct passive form: "The chef is cooking dinner."
- Rearrange: 1) has become 2) a necessity 3) in modern life 4) the internet
- Best inference from: "Although the startup raised significant funding, it shut down within a year due to poor market fit."  
(a) Funding guarantees success (b) Market fit matters more than funding alone (c) The founders were inexperienced (d) The investors made a mistake

# PART 9 — Mock Test Answer Key & Explanations

## Section I: Quantitative Aptitude

| #  | Answer          | Brief Explanation                                                                                                                                                                                                                         |
|----|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | 6               | Unit digit of $6^{\text{anything}}$ =6. Unit digit of $7^{92}$ : cycle(7)=7,9,3,1; $92 \bmod 4=0 \rightarrow 4\text{th term}=1$ . $6 \times 1=6$                                                                                          |
| 2  | 84              | Other number = $(\text{HCF} \times \text{LCM})/48 = (12 \times 336)/48 = 84$                                                                                                                                                              |
| 3  | 6.25%           | SI for 2 yrs (5yr amt – 3yr amt) = 600 $\rightarrow$ SI/yr=300. Principal = $5700 - (3 \times 300) = 4800$ . Rate = $(\text{SI} \times 100)/(P \times T) = (300 \times 100)/4800 = 6.25\%$                                                |
| 4  | Rs.667 (approx) | $\text{SP}=1.2 \times 500=600$ . $\text{MP}(1-0.10)=600 \rightarrow \text{MP}=600/0.9=666.67$                                                                                                                                             |
| 5  | 9 years         | Present ages $A=3k$ , $B=5k$ . After 9 yrs: $(3k+9):(5k+9)=3:4 \rightarrow 4(3k+9)=3(5k+9) \rightarrow 12k+36=15k+27 \rightarrow 3k=9 \rightarrow k=3$ . A's present age = $3k = 9$ years (check: after 9 yrs, $18:24 = 3:4 \checkmark$ ) |
| 6  | 20 workers      | $15 \times 24=360$ man-days. $360/18=20$                                                                                                                                                                                                  |
| 7  | 2 hours         | Downstream speed= $12+4=16\text{km/hr}$ . Time= $32/16=2$ hrs                                                                                                                                                                             |
| 8  | 120 ways        | $5!=120$                                                                                                                                                                                                                                  |
| 9  | 3/8             | Total=8, green=3. $P=3/8$                                                                                                                                                                                                                 |
| 10 | 29              | $a=2, d=3$ . $T_{10}=2+9 \times 3=29$                                                                                                                                                                                                     |
| 11 | 54              | Heron's: $s=18$ ; Area= $\sqrt{(18 \times 9 \times 6 \times 3)}=\sqrt{2916}=54$                                                                                                                                                           |
| 12 | 18:5            | $x=3k, y=4k$ . $(6k+12k):(9k-4k)=18k:5k=18:5$                                                                                                                                                                                             |
| 13 | 42              | Total of 7 numbers = $7 \times 24=168$ . Total of remaining 6 = $6 \times 21=126$ . Excluded number = $168-126 = 42$                                                                                                                      |
| 14 | 0.65            | Form $a^3-b^3$ over $a^2+ab+b^2 = (a-b)$ where $a=0.75, b=0.1 \rightarrow 0.65$                                                                                                                                                           |
| 15 | 5%              | $(1+r/100)^3=9261/8000=1.157625=1.05^3 \rightarrow r=5\%$                                                                                                                                                                                 |

## Section II: Logical Reasoning

| #  | Answer        | Brief Explanation                                                                                                                                                      |
|----|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 16 | 36            | Perfect squares: $6^2=36$                                                                                                                                              |
| 17 | DIBJS         | Each letter +1: $C \rightarrow D, H \rightarrow I, A \rightarrow B, I \rightarrow J, R \rightarrow S$                                                                  |
| 18 | Son           | "My father's only daughter" = Rita herself (assuming she's the only daughter). So the boy is Rita's own son.                                                           |
| 19 | 10 km         | $\sqrt{(6^2+8^2)}=\sqrt{100}=10$ (Pythagoras)                                                                                                                          |
| 20 | Valid         | Transitive chain: square $\subset$ rectangle $\subset$ quadrilateral                                                                                                   |
| 21 | G is leftmost | Order derived: G-F-E-D (G left of F, F left of E, E left of D)                                                                                                         |
| 22 | $3+1+2=6$     | $C=3, A=1, B=2 \rightarrow \text{sum}=6$                                                                                                                               |
| 23 | 5 Sundays     | Starts Sunday, 31 days $\rightarrow$ Sundays on 1,8,15,22,29 = 5                                                                                                       |
| 24 | Grandmother   | D is A's mother = B's grandmother                                                                                                                                      |
| 25 | 200           | Others are perfect squares ( $11^2, 12^2, 13^2, 15^2$ ); 200 is not                                                                                                    |
| 26 | 112           | Pattern: $\times 2$ each time: $56 \times 2=112$                                                                                                                       |
| 27 | QFODJM        | Each letter shifted +1 (as shown by PAPER $\rightarrow$ QBQFS): $P \rightarrow Q, E \rightarrow F, N \rightarrow O, C \rightarrow D, I \rightarrow J, L \rightarrow M$ |
| 28 | Monday        | 29th is $(29-1)=28$ days after the 1st. $28 \bmod 7 = 0$ , so it falls on the same weekday as the 1st $\rightarrow$ Monday                                             |
| 29 | Strong        | Directly relevant, protects third-party health — valid strong argument                                                                                                 |
| 30 | 34            | Total = $9+26-1 = 34$ students                                                                                                                                         |

## Section III: Verbal Ability

| # | Answer |  |
|---|--------|--|
|---|--------|--|

|    |                                                                 | Brief Explanation                                                             |
|----|-----------------------------------------------------------------|-------------------------------------------------------------------------------|
| 31 | (b) Frank                                                       | Candid = honest and direct                                                    |
| 32 | (b) Lazy                                                        | Diligent = hardworking; opposite = lazy                                       |
| 33 | are                                                             | "Neither...nor" agrees with the nearer subject (employees, plural) → "are"    |
| 34 | (b) Receive                                                     | "i before e except after c" rule                                              |
| 35 | "friend" → "friends" / "are" → "is"                             | "One of my friends is getting married" — fix both the noun and verb agreement |
| 36 | (a) Bilingual                                                   | Bi- = two, lingual = language                                                 |
| 37 | (b) Avoid the main topic                                        | Standard idiom meaning                                                        |
| 38 | "Dinner is being cooked by the chef."                           | Passive voice: continuous tense structure "is being + V3"                     |
| 39 | 4,2,3,1 → "The internet has become a necessity in modern life." | Logical subject-verb-object order                                             |
| 40 | (b)                                                             | Directly supported inference without overreach                                |

**Scoring guide:** 34+/40 = Excellent, ready for most company-level tests. 26-33 = Good, revise weak topics from the explanations above. Below 26 = Revisit Part 1-7 topic-wise before re-attempting.

— End of Volume 2 — Use alongside the Master Aptitude Prep Guide (Volume 1) for complete topic + practice coverage. —